

Periodic Table of the Elements

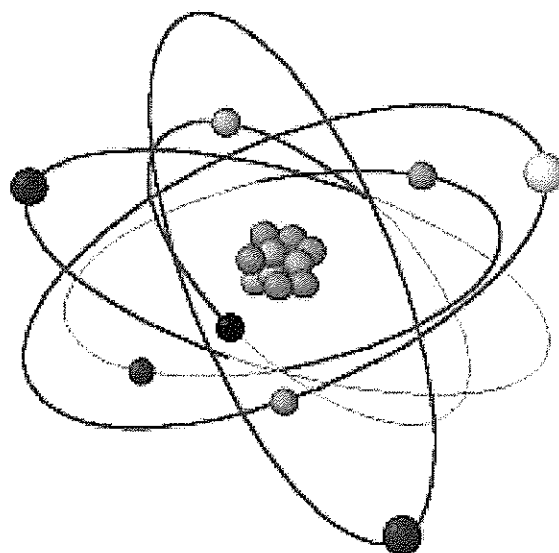
1	H	2	He																	18	Ar	19	K	20	Ca	21	Sc	22	Ti	23	V	24	Cr	25	Mn	26	Fe	27	Co	28	Ni	29	Cu	30	Zn	31	Ga	32	Ge	33	As	34	Se	35	Br	36	Kr
3	Li	4	Be																	18	Ar	19	K	20	Ca	21	Sc	22	Ti	23	V	24	Cr	25	Mn	26	Fe	27	Co	28	Ni	29	Cu	30	Zn	31	Ga	32	Ge	33	As	34	Se	35	Br	36	Kr
4	Na	5	Mg																	18	Ar	19	K	20	Ca	21	Sc	22	Ti	23	V	24	Cr	25	Mn	26	Fe	27	Co	28	Ni	29	Cu	30	Zn	31	Ga	32	Ge	33	As	34	Se	35	Br	36	Kr
5	K	6	Ca	7	Sc	8	Ti	9	V	10	Cr	11	Mn	12	Fe	13	Co	14	Ni	15	Cu	16	Zn	17	Ga	18	Ge	19	As	20	Se	21	Br	22	Kr																						
6	Rb	7	Sr	8	Y	9	Zr	10	Nb	11	Mo	12	Tc	13	Ru	14	Rh	15	Pd	16	Ag	17	Cd	18	In	19	Sn	20	Sb	21	Te	22	I	23	Xe																						
7	Cs	8	Ba	9	La	10	Hf	11	Ta	12	W	13	Re	14	Os	15	Ir	16	Pt	17	Au	18	Hg	19	Tl	20	Pb	21	Bi	22	Po	23	At	24	Rn																						
8	Fr	9	Ra	10	Ac	11	Th	12	Pa	13	U	14	Np	15	Pu	16	Am	17	Cm	18	Bk	19	Cf	20	Es	21	Fm	22	Md	23	No	24	Lr																								

• Lanthanide Series

57	Ce	58	Pr	59	Nd	60	Pm	61	Sm	62	Eu	63	Gd	64	Tb	65	Dy	66	Ho	67	Er	68	Tm	69	Yb	70	Lu
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• Actinide Series

89	Th	90	Pa	91	U	92	Np	93	Pu	94	Am	95	Cm	96	Bk	97	Cf	98	Es	99	Fm	100	Md	101	No	102	Lr
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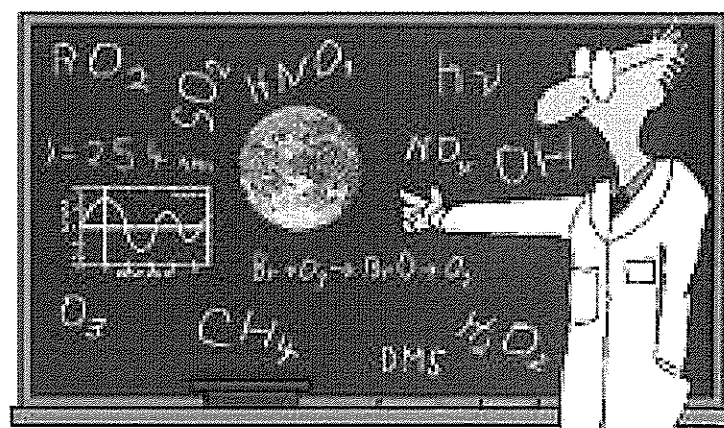
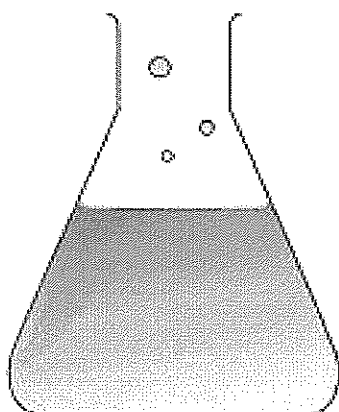


YEAR 10 SCIENCE

NATURAL & PROCESSED MATERIALS

CHEMISTRY

Name: _____



YEAR 10 SCIENCE
NATURAL AND PROCESSED MATERIALS
CHEMICAL BONDING - GLOSSARY

Atom - fundamental building block of matter.

Proton - positively charged particle found in the nucleus of an atom.

Neutron - neutral particle found in the nucleus of an atom.

Electron - negatively charged particle found around the nucleus; is electrostatically attracted to the protons in the nucleus.

Element - substance made up of atoms that are all the same (i.e. same atomic number).

Atomic Number - number of protons in the nucleus.

Atomic Mass - the number of protons and neutrons in the nucleus.

Molecule - two or more atoms chemically combined.

Compound - two or more elements chemically combined in fixed proportions.

Periodic Table - arranges elements in order of increasing atomic number but groups them according to their chemical properties.

Period - a row across the periodic table.

Group - a column going down the periodic table; all elements in a group have the same chemical properties.

Metals - found on the left hand side of the periodic table; donate electrons during reactions.

Malleable - able to be beaten into thin sheets.

Ductile - able to be made (drawn) into a wire.

Lustre - the shine of a metal.

Alloy - a mixture of two or more metals.

Metallic Bond - electrostatic attraction between positive metal ions and delocalised electrons.

Non-metals - found on the right hand side of the periodic table; accept electrons during reactions.

Alkali Metals - belong to Group I of the periodic table; react with water to form alkali (basic) solutions.

Alkali Earth Metals - belong to Group II of the periodic table; not as reactive as the alkali metals.

Transition Metals - found in the middle of the periodic table; most reactive metals are at the top.

Halogens - belong to Group VII of the periodic table; form salts when reacted with metals.

Noble (Inert) Gases - belong to Group VIII of the periodic table; do not react with other elements.

Allotrope - a different form of the same element.

Ion - an atom (or group of atoms) that has a charge due to the gain or loss of electrons.

Valency - the charge on an ion.

Ionic Bond - the electrostatic attraction between oppositely charged ions.

Covalent Bond - formed by the sharing of two electrons between two atoms.

Covalent Network - atoms bond to each other in a 3-dimensional arrangement; elements in Group IV do this.

Solute - substance that dissolves.

Solvent - the substance that does the dissolving.

Solution - formed when one substance dissolves into another.

Precipitate - a solid that forms when two ionic solutions are mixed, forming an insoluble substance.

Spectator Ions - not directly involved in forming a precipitate.

Ionic Equation - shows only the ions involved in a reaction.

ELEMENTS AND COMPOUNDS

Definitions

Element: substance that contains atoms that are all the same.

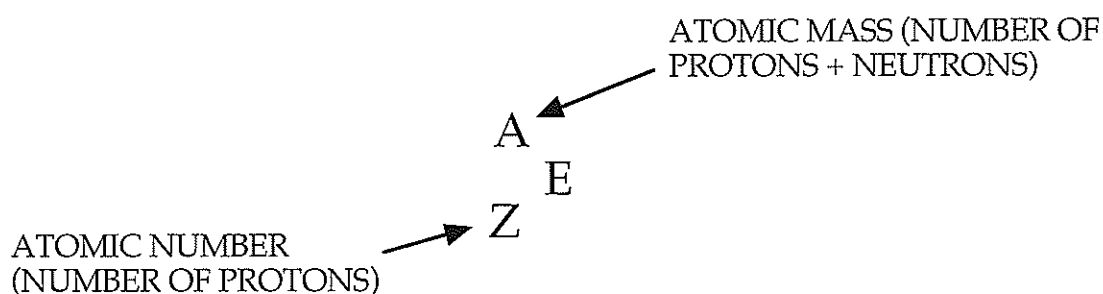
(Elements have atoms with the same number of protons in the nucleus.)

Atomic Number: number of protons in the nucleus.

Mass Number: number of protons and neutrons in the nucleus.

Each element has its own particular number of protons, neutrons and electrons.

These numbers are represented using the following general symbol.



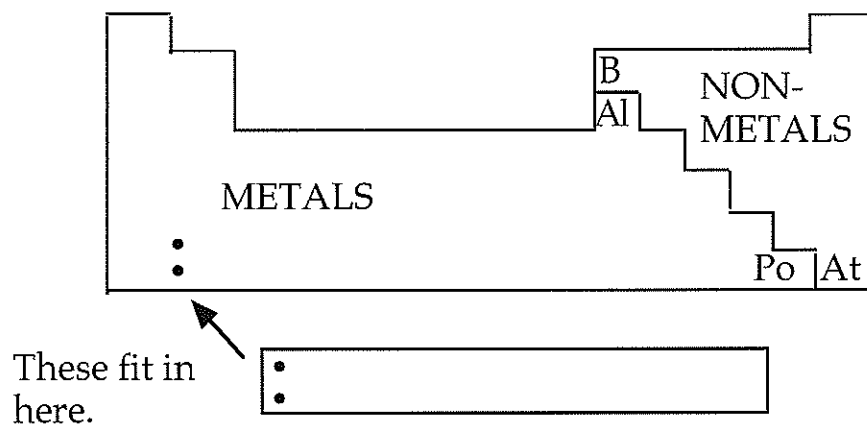
Examples

ELEMENT	PROTONS	NEUTRONS	ELECTRONS
$^{24}_{12}\text{Mg}$	12	12	12
$^{23}_{11}\text{Na}$	11	12	11
$^{238}_{92}\text{U}$	92	146	92

There are 92 naturally-occurring elements and 22 man-made ones.

These are arranged by atomic number in the Periodic Table.

The elements can be classified as metals and non-metals.



Metals are found on the left hand side of the table, non-metals to the right.

Along the diagonal shown are *metalloids* or *semi-metals* that show properties of both metals and non-metals.

The symbols for elements always start with a capital letter, and the second letter (if used) is always lower case.

e.g. Al - aluminium C - carbon

The symbol is based on the English name, though some are based on Latin, Arabic and German names.

NON-ENGLISH		ENGLISH	
sodium	Na	hydrogen	H
potassium	K	helium	He
iron	Fe	lithium	Li
copper	Cu	beryllium	Be
silver	Ag	boron	B
tungsten	W	carbon	C
gold	Au	nitrogen	N
mercury	Hg	oxygen	O
lead	Pb	fluorine	F
tin	Sn	neon	Ne
		magnesium	Mg
cobalt	Co	aluminium	Al
manganese	Mn	silicon	Si
bromine	Br	phosphorus	P
iodine	I	sulfur	S
zinc	Zn	chlorine	Cl
nickel	Ni	argon	Ar
barium	Ba	calcium	Ca

PERIODIC TABLE

PERIODIC TABLE																	
1 H 1.01																	
3 Li 6.94	4 Be 9.01																
11 Na 23.0	12 Mg 24.3	13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 39.9										
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.4	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc 98.0	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71 See below	72 Hf 178.5	73 Ta 180.9	74 W 183.0	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po 209	85 At 210	86 Rn 222

Table of Salt Solubilities

RULE	EXCEPTION
ALL SALTS OF K ⁺ , Na ⁺ and NH ₄ ⁺ are SOLUBLE	
ALL NITRATE compounds are SOLUBLE	
ALL ACETATE compounds are SOLUBLE	
ALL CHLORIDE compounds are SOLUBLE	AgCl, CuCl, PbCl ₂
ALL SULPHATE compounds are SOLUBLE	Ag ₂ SO ₄ , BaSO ₄ , CaSO ₄ , PbSO ₄
ALL SULPHIDE compounds are INSOLUBLE	HgS, CuS, K ₂ S, HgS, Na ₂ S, (NH ₄) ₂ S
ALL CARBONATE compounds are INSOLUBLE	K ₂ CO ₃ , Na ₂ CO ₃ , (NH ₄) ₂ CO ₃
ALL PHOSPHATE compounds are INSOLUBLE	K ₃ PO ₄ , Na ₃ PO ₄ , (NH ₄) ₃ PO ₄
ALL OXIDES, HYDROXIDES, K ⁺ , Na ⁺ etc. INSOLUBLE	

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm 145	62 Sm 150.4	63 Eu 152.0	64 Gd 157.2	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0
89 Ac 227.0	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np 237.0	94 Pu 244	95 Am 243	96 Cm 247	97 Bk 247	98 Cf 251	99 Es 252	100 Fm 257	101 Md 258	102 No 259

6 ← Atomic Number
C ← Symbol
12.0 ← Relative Atomic Mass (Atomic Weight)

ACTIVITY SERIES

K Na Ca Mg Al Zn Fe Ni Sn Pb H Cu Hg Ag Pt Au

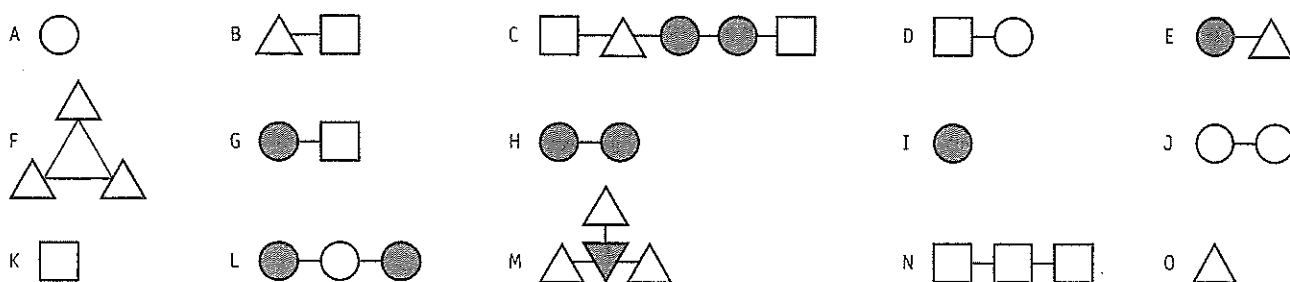
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Matter can be divided into groups using certain criteria. Matter can be divided into living and non-living things. Matter can be divided into solids, liquids and gases. Matter can be grouped into metals and non-metals. Matter can be split into single particles and multiple particles. Matter can be divided into substances which contain one kind of atom or more than one kind of atom.

Definitions

- **Atoms** are tiny particles. They are the simplest form of matter. Atoms are made up of single particles.
- **Molecules** are also tiny particles. They are made up of two or more atoms. These atoms are held together by chemical bonds.
- **Elements** are substances which contain only one kind of atom.
- **Compounds** are substances which contain particles that are made up of two or more different atoms.

Look at the following diagrams showing various particles and substances. Then answer the questions below by referring to the definitions.



1 Identify all atoms and all molecules and record them in the table.

Particles	Atoms	Molecules
Diagrams		

2 Identify all elements and all compounds and record them in the table.

Substances	Elements	Compounds
Diagrams		

3 Complete the following table about particles and substances by adding as many examples of each as possible from the diagrams.

Atom of element	Atom of compound	Molecule of element	Molecule of compound

4 Explain why one cell in the table from question 3 is left empty. _____

5 Make up two more examples for each cell in question 3 and draw them below the table.

6 Explain what the lines in the diagrams represent. _____

4



+

Archimedes of Syracuse (287 – 212 BC) ran naked out into the streets crying 'Eureka!', excited after finding out the answer to King Hieron II's problem.

1 The following words relate to this topic about matter but are incomplete. Identify the missing letters and then use them and their matching numbers to complete the sentence below and discover the problem King Hieron had.

ma	te		ato	s		s	ubstan	e		mol		ule			a	tic	es
1	6		2			3		5		4	5	8			17	6	15

o	l	d	b	n	s	o	r		r	i	t	e	r	i		m	t	a	s
8	7		11	13		9	2		5				10		4		15		

l qu d s lid g s co poun s ater
7 7 11 10 2 13 12

sil__e__ __ydr__en __le__e__t
14 6 16 11 19 4 2 18

King Hieron II's problem

13713	1164	191115138271168	384	101515	1164	19111513	116	6417151054
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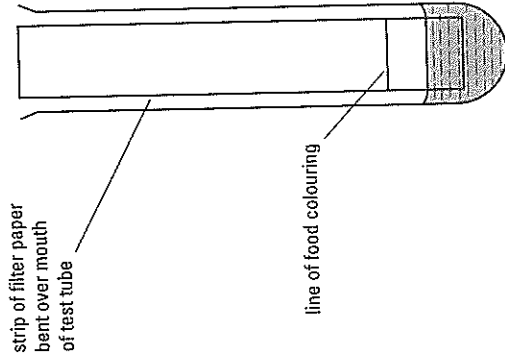
[illegible]

2 Write each of the words listed above into a sentence about science.

[illegible]

3 Research: Use library resources to find out how Archimedes solved the king's problem. This should explain why he was naked at the time!

Fig. 3.1



Processing of results, and questions

- 1 Identify those food colourings or dyes which consisted of a single component, and those which contained more than one component.
- 2 What property of the food colourings enables them to be separated in this way?

Experiment

4 Empirical formula of magnesium oxide

Background

The empirical formula of a compound is the simplest value of the ratio of atoms of each element in the compound. In this experiment you will determine the empirical formula of magnesium oxide.

Equipment required

Balance
Crucible and lid
Bunsen
Magnesium ribbon (0.2 g, about 20 cm)
Steel wool
Crucible tongs
Pipe clay triangle
Tripod

Procedure

#1 Draw up a table for your results as shown below:

Mass of crucible and lid	=	g
Mass of crucible, lid and magnesium	=	g
Mass of magnesium	=	g
Mass of crucible, lid and magnesium oxide	=	g
Mass of magnesium oxide	=	g
Mass of oxygen combined with magnesium	=	g

#2 Obtain a clean, dry crucible and lid and heat them for 5 minutes over a bunsen flame. Allow them to cool.

#3 Thoroughly clean the surfaces of a 20 cm strip of magnesium ribbon with steel wool.

#4 Coil the magnesium ribbon so that it will fit into the crucible.

#5 Accurately weigh the crucible and lid on a balance and record the mass in your results table.

#6 Place the magnesium ribbon into the crucible, replace the lid and reweigh.

Experiment

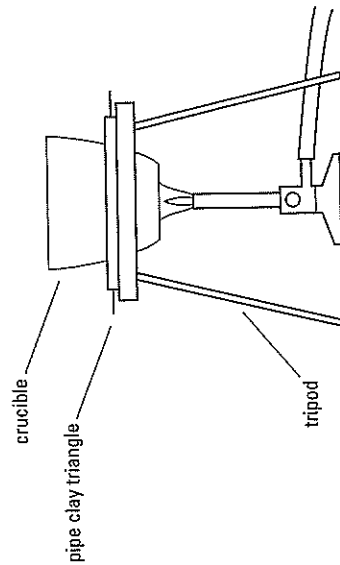
5 Water of crystallisation of barium chloride

Safety Note

- Care must be taken when burning magnesium ribbon.
- Do not touch the hot crucible with your hands. Always use crucible tongs.

- #7 Carefully heat the crucible and its contents with the lid off until the magnesium begins to glow. Immediately replace the lid and heat the crucible strongly.

Fig. 4.1



- #8 Continue to heat the crucible for about 10 minutes, occasionally lifting the lid with tongs to provide oxygen for the reaction. Try to avoid the loss of any magnesium oxide smoke when the lid is lifted.
- #9 When all the magnesium has reacted, remove the lid and heat strongly for a further 5 minutes.
- #10 Replace the crucible lid and allow the crucible and contents to cool.
- #11 Reweigh the crucible with its contents and lid.

Processing of results, and questions

- From your results table calculate the
 - mass of magnesium which reacted,
 - mass of oxygen combined with the magnesium.
- Calculate the numbers of moles of magnesium and oxygen and the ratio $n(\text{Mg}) : n(\text{O})$. What is the empirical formula of magnesium oxide?
- Why are the crucible and lid heated at the beginning of the experiment before being weighed?
- What are the possible sources of error in this experiment?

Background

Many ionic compounds which crystallise out of an aqueous solution have water molecules fixed in regular positions throughout the crystal lattice. These compounds are said to be *hydrated*, and the number of water molecules per formula unit of the compound is shown as part of the formula.

For example $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ have respectively five and ten water molecules of crystallisation per formula unit of compound.

These water molecules of crystallisation can be driven off by heating the hydrated compound. If the remaining compound has no water molecules of crystallisation, it is said to be *anhydrous*.

The purpose of this experiment is to determine the number of water molecules of crystallisation of hydrated barium chloride, $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$. To do this, a known mass of the hydrated compound will be heated so that the mass of water driven off can be determined. From this the empirical formula of the hydrate will be calculated.

Equipment required

Balance
Crucible and lid
Bunsen
Hydrated barium chloride (2–3 g)
Pipe clay triangle
Tripod
Desiccator
Crucible tongs

Procedure

Safety Note

- Do not touch the hot crucible with your hands. Always use crucible tongs.

- #1 Obtain a clean, dry crucible and lid and heat them strongly for 5 minutes over a bunsen flame. Allow them to cool in a desiccator.

Science inquiry

Verbal/Linguistic Visual/Spatial Logical/Mathematical

Before chemists understood the atomic theory of matter, people thought that the process of combustion involved the release of a material they called 'phlogiston'. Today, scientists know that phlogiston does not exist. Instead, combustion occurs when chemicals react vigorously with oxygen.

vigorously (adv) with a lot of energy

failing (n) fault; problem

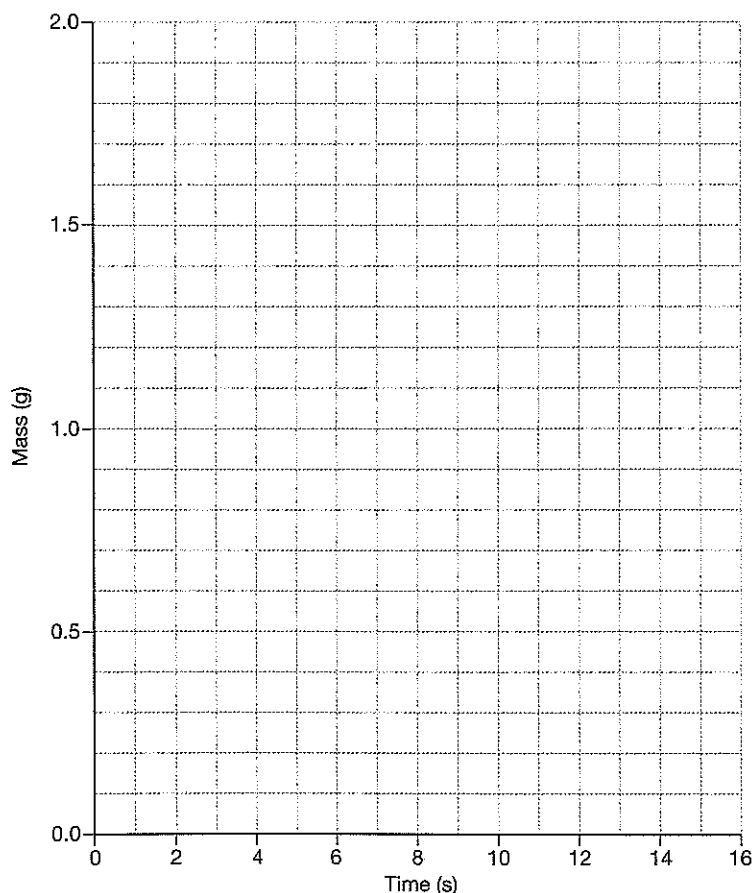
One of the failings of the phlogiston theory was that it could not explain the combustion of metals. The phlogiston theory predicted that everything that combusts should decrease in mass, due to the release of phlogiston. However, when metals burn they increase in mass. This is because they combine with oxygen to form a metal oxide, which is heavier than the metal.

Zac designed an experiment to measure the mass of 1 g of magnesium (Mg) as it burnt in oxygen (O_2) to produce magnesium oxide (MgO). His results are summarised in the tables below.

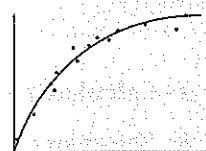
Time (s)	0	1	2	3	4	5	6	7	8
Mass (g)	1.00	1.04	1.08	1.12	1.16	1.20	1.24	1.29	1.34

Time (s)	9	10	11	12	13	14	15	16
Mass (g)	1.39	1.44	1.49	1.55	1.61	1.67	1.67	1.67

- 1 Plot the data points on the axes below to **construct** a graph. Draw a line of best fit.

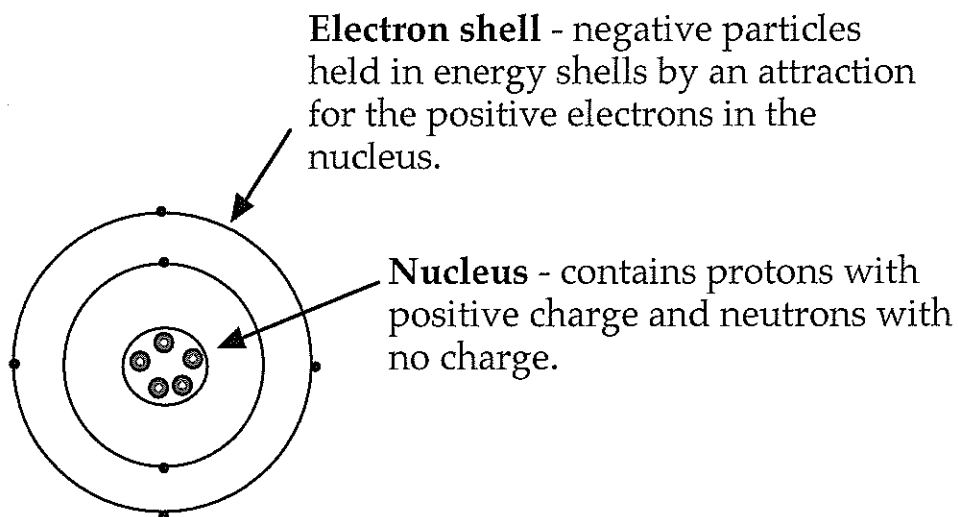
**HINT**

A 'line of best fit' is a smooth line that goes near most of the points. For example:



ATOMIC STRUCTURE

Bohr Model of the Atom



Element- contains only one type of atom.

Compound - two or more elements chemically combined in fixed proportions.

Molecule - two or more atoms chemically combined.

Energy Shells - electrons are found in distinct energy shells around the nucleus.

Each of the energy shells has a maximum number of electrons that can be held.

WE WILL CONSIDER THE FIRST 20 ELEMENTS ONLY.

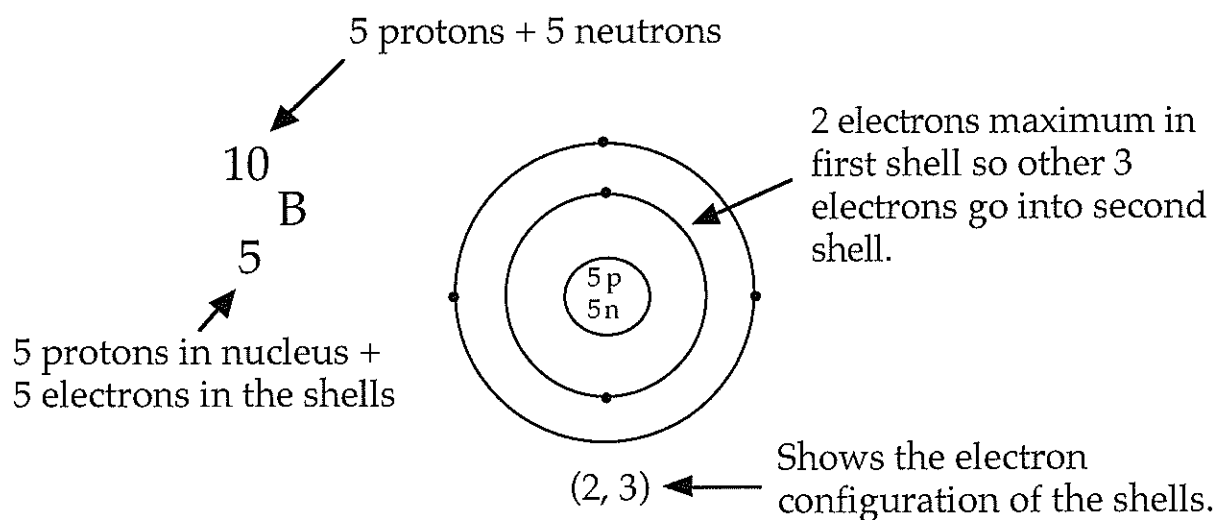
1 st shell	2
2 nd shell	8
3 rd shell	8
4 th shell	2

The arrangement of electrons in the shells is called the ***electron configuration*** for the element.

It is possible to draw diagrams to show the electron configuration of elements.

The Periodic Table is very handy for this!

Examples



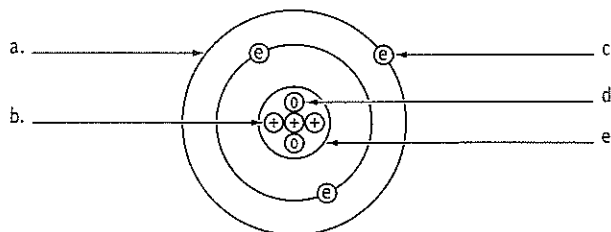
35 Cl
17

Note: You will need to refer to a copy of the periodic table for this worksheet.

A popular atomic model

Many people today believe that atoms can be described by the atomic model of the solar system atom. This orbital atomic model allows us to explain many processes.

The atom is made up of three main subatomic particles which are grouped into two lots. Protons and neutrons are tightly packed into the nucleus which forms the centre of the atom. Electrons surround the nucleus on several orbits (or shells).



1 Label the diagram of an atomic model (lithium).

Subatomic particles

Protons and neutrons are 2000 times heavier than electrons. Protons and electrons have an electric charge: protons are positive and electrons are negative. Electrons fly around the nucleus on orbits. Neutrons and protons rest in the nucleus.

2 Complete the table about subatomic particles.

Name	Charge	Mass	Position	Movement
			on orbits	
		2000 times		
	neutral			no

Atomic numbers and mass numbers, atoms and isotopes

Atoms are neutral, although some of their subatomic particles do have an electric charge. But the positive and negative subatomic charges cancel each other leaving the atom neutral. Therefore, the number of positive protons must equal the number of negative electrons. The *atomic number* gives the number of protons.

Since protons and neutrons are very much heavier than electrons, it is the protons and neutrons which determine the mass of the atoms. Usually an atom has as many neutrons as protons. The *mass number* is the sum of protons and neutrons.

Some elements have atoms which have the same number of protons in the nucleus (so the same atomic number) but more or fewer neutrons (so they have different atomic weights). Their atoms are now called *isotopes*.

When the number of neutrons is much larger than the number of protons in a large nucleus, the isotope's nucleus is unstable. As a result it breaks up, forming new atoms and giving off radiation.

3 Complete the table about mass and atomic numbers of atoms.

Name	Mass number	Atomic number	Number of protons	Number of neutrons	Number of electrons	Scientific notation
lithium	5	3	3	2	3	${}^5_3\text{Li}$
magnesium	24				12	
aluminium		13		13		
hydrogen	1					
chlorine						${}^{35}_{17}\text{Cl}$
phosphorus						${}^{31}_{15}\text{P}$
oxygen						${}^{16}_8\text{O}$



+

Electrons on shells

Electrons move on orbits (or shells) around the nucleus. The inside shells are smaller than the outer shells and so can hold fewer electrons. Inner shells are filled up before the outer shells. The first innermost shell can carry a maximum of two electrons, the second and third shells can hold no more than eight electrons (these maximum numbers are true for the first 20 atoms).

- 4** Draw atomic models for the following atoms. The mass and atomic numbers give you a clue to the number of protons, neutrons and electrons.

helium (${}^4_2\text{He}$)

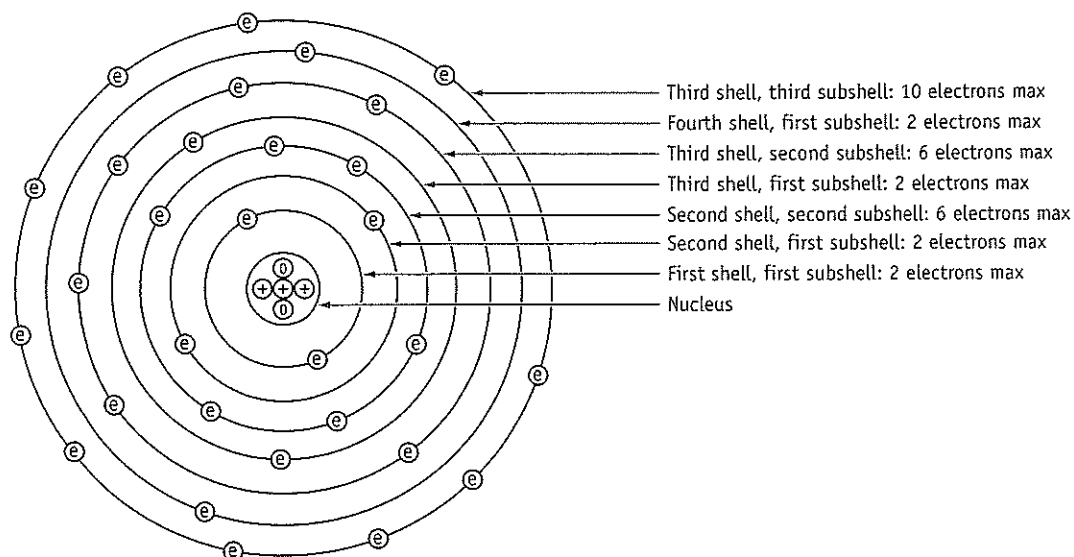
silicon (${}^{28}_{14}\text{Si}$)

beryllium (${}^9_4\text{Be}$)

potassium (${}^{39}_{19}\text{K}$)

Shells and their subshells

The orbital model of the atom becomes untidy with the twenty-first atom, because the electron shells are actually composed of up to four subshells. And some of the subshells overlap each other. The third subshell of the third shell is actually bigger than the first subshell of the fourth shell.



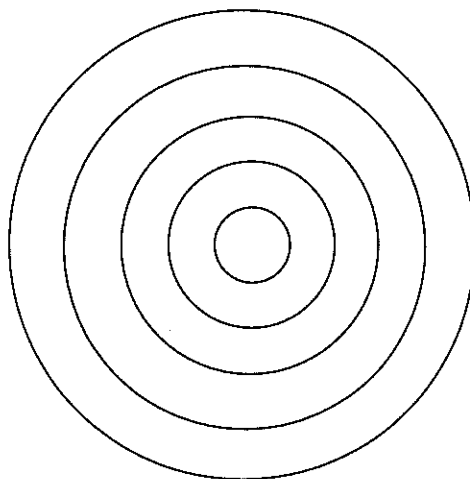
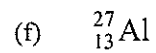
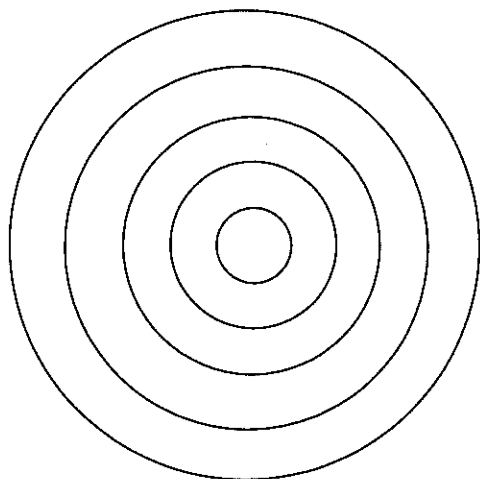
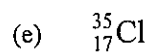
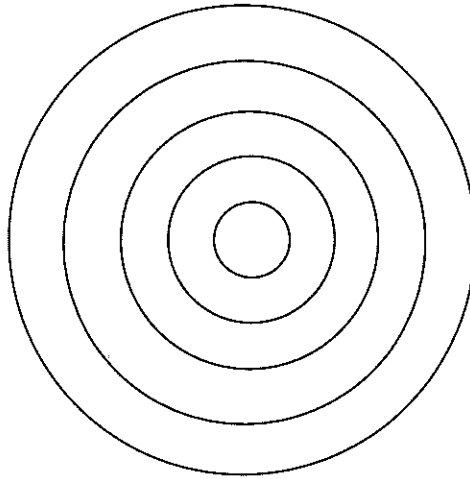
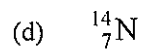
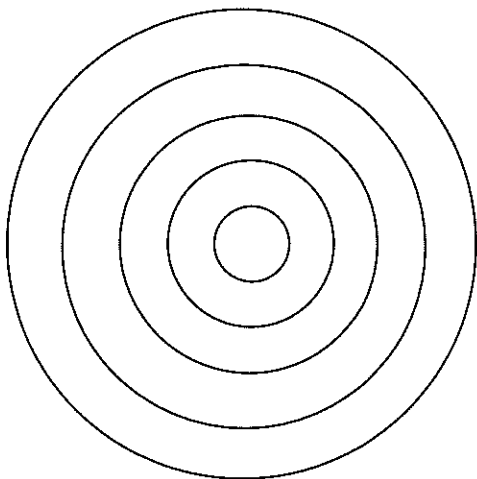
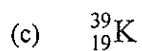
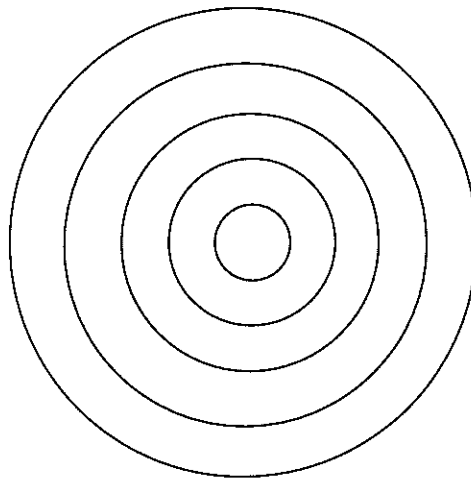
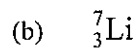
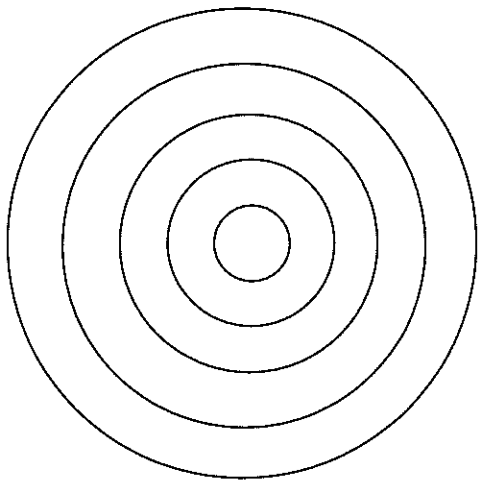
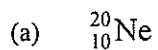
- 5** State the number of protons, neutrons and electrons for the atom of iron (${}^{56}_{26}\text{Fe}$).

- 6** Shade in electrons to the atomic model above for iron.

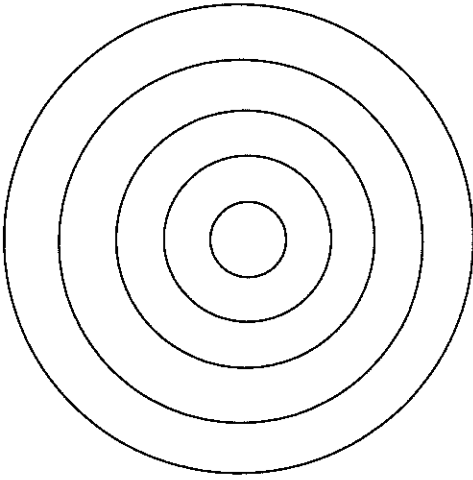
- 7** State the number of subshells for the first three shells. _____

YEAR 9 SCIENCE
NATURAL AND PROCESSED MATERIALS
WORKSHEET - ELECTRON CONFIGURATION DRAWINGS

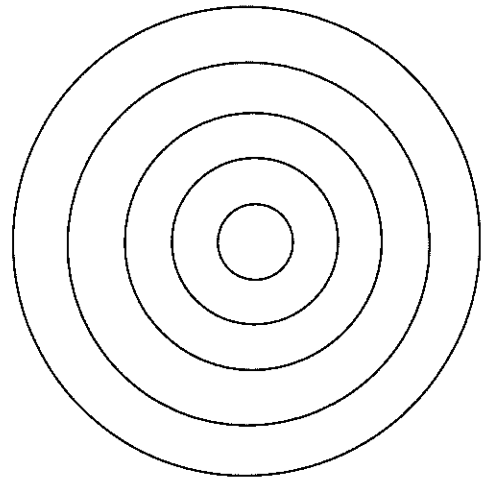
Draw electron configuration diagrams for the following elements. Show the *number of protons and neutrons* in the nucleus, and the *number of electrons* in each shell.



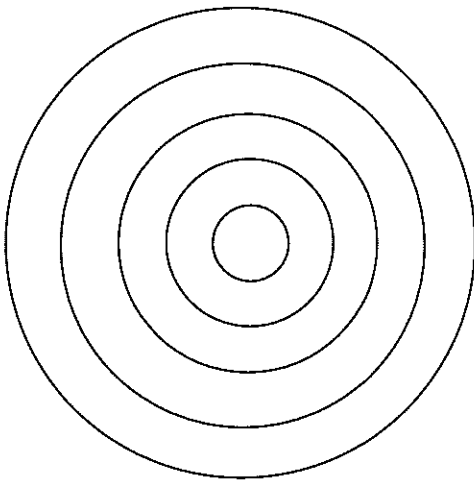
(g) $^{12}_6\text{C}$



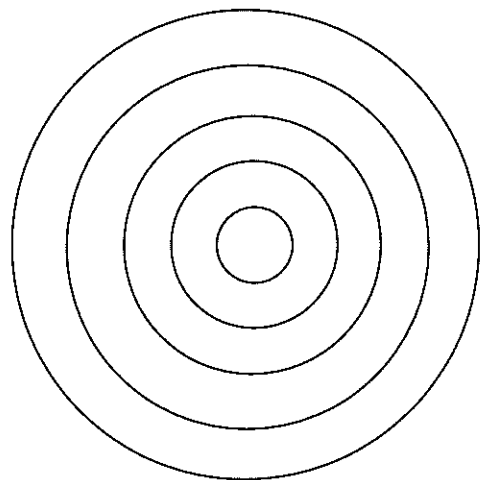
(h) ^4_2He



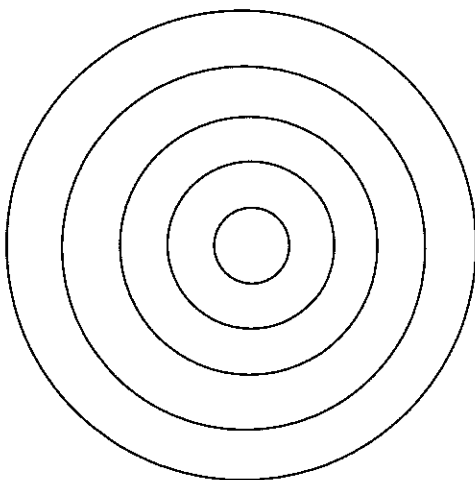
(i) $^{32}_{16}\text{S}$



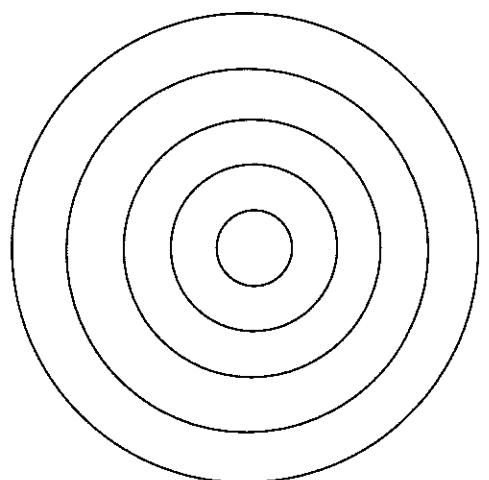
(j) $^{31}_{15}\text{P}$



(k) $^{10}_5\text{B}$



(l) $^{40}_{20}\text{Ca}$



Around the chemical maze

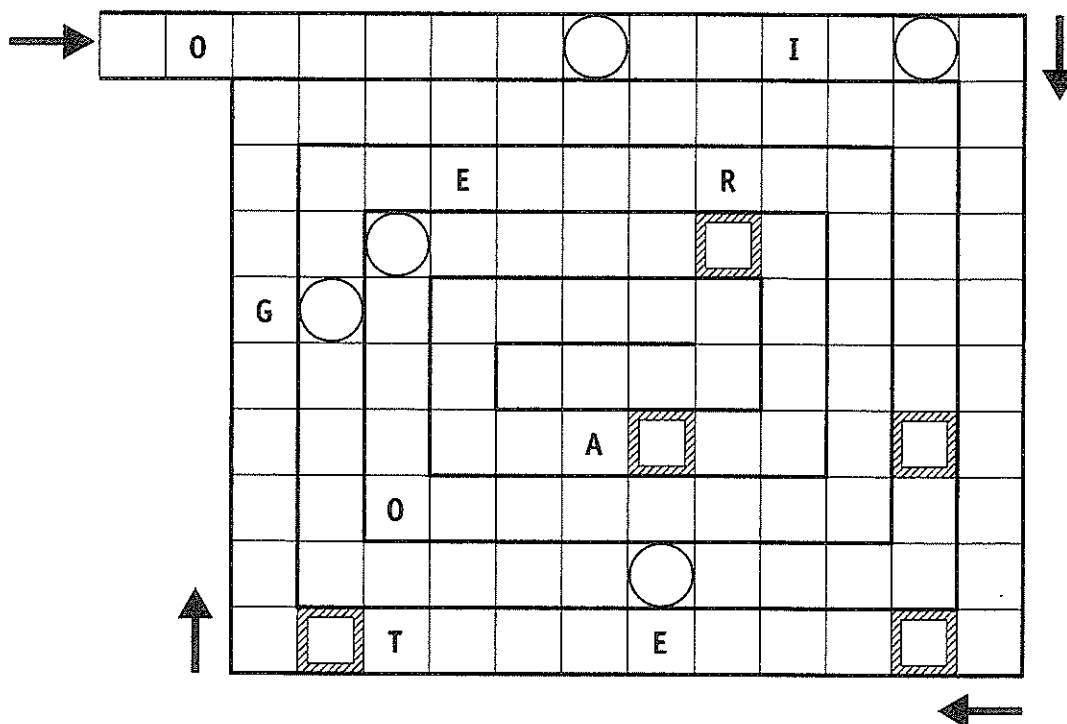
9

+

Find out who I am by following the instructions below.

In 1566 I was involved in a duel with another student and as a result lost part of my nose. For the rest of my life I wore a metal insert over the missing part. I was involved in the study of heavenly bodies.

Read through the clues and write the answers into the maze, beginning at the arrow at the top left corner. The last letter of each answer is the first letter to the next answer. Some letters are already supplied in the maze. Unjumble the underlined letters to reveal Who I Am.



Clues

- 1 Metal with the symbol Na.
- 2 Shiny malleable substance.
- 3 Element with three protons.
- 4 Able to be pressed or bent into shape.
- 5 Simple pure substance that cannot be broken down into anything simpler.
- 6 Metal with 22 protons.
- 7 Metal with symbol Mg.
- 8 Solids, liquids and gases are all states of _____.
- 9 Noble gas in column 8 of the periodic table.
- 10 Gas in our atmosphere.
- 11 Metal with symbol Ni.
- 12 Water at room temperature.
- 13 Scientist who noticed that lithium, sodium and potassium had very similar properties.
- 14 Dimitri Mendeleev came from which country?
- 15 Metal with symbol Al.
- 16 Scientist who resolved some problems on the periodic table by basing the order on atomic number instead of weight.
- 17 One metal in the periodic table that starts with Y.
- 18 Compound with the formula CH_4 .
- 19 Element with symbol Es.

Who am I?

The circled letters form my given name and the boxed letters my family name. _____

Science as a human endeavour

Verbal/Linguistic

Many scientists contributed to our understanding of the elements, their properties and the family relationships between them. Each of these scientists therefore contributed to the development of the periodic table we use today. Some of these scientists are discussed in the Science as a Human Endeavour on pages 115–16 of your student book. This table provides some more information about them.

Scientist	Lived	Scientific contributions	Other facts
Antoine-Laurent de Lavoisier	1743–93	• Did not discover any new elements • Identified and named the existing 33 elements • Proved that mass was not destroyed in a reaction • Proved that rusting involved oxygen bonding to iron (increasing its mass) • Proved that combustion needed oxygen	• Nationality: French • Was very wealthy; worked as a tax collector • Had 13 000 beakers in his laboratory • Was executed by guillotine after the French Revolution • His statue in Paris was made with a 'spare' head from another statue
John Dalton	1766–1844	• Revived the Ancient Greek idea of atoms • Gave each element a pictorial symbol, which is not used today • Predicted atomic masses • Developed laws that explain how gases behave • Showed that compounds were made of elements in definite mass proportions	• Nationality: English • Came from a very poor family • Was put in charge of his small school when he was 12 years old • Still taught mathematics at a primary school when he was old and famous
Jons Jacob Berzelius	1779–1848	• Used atomic mass to arrange the elements • Gave the elements symbols that we still use today, based on their Greek or Latin names • Showed that compounds were made of elements in definite mass proportions • Was the first to classify substances as organic (containing C) or inorganic (no C)	• Nationality: Swedish • Used superscript numbers instead of subscript numbers in chemical formulas—wrote water as H²O instead of H₂O
Johann Dobereiner	1780–1849	• Placed elements into 'triads' (families of three)	• Nationality: German



Scientist	Lived	Scientific contributions	Other facts
John Newlands	1837–98	- Used atomic mass to arrange elements - Noticed every eighth element 'harmonising' (forming patterns) - Placed elements in periods of 7 elements	- Nationality: English - Called his arrangement 'the law of octaves' after musical octaves
Lothar Meyer	1830–95	- Constructed a table very similar to Mendeleev's - Finished his periodic table before Mendeleev - Published his table after Mendeleev	- Nationality: German - Is regarded as a 'father' of the periodic table
Dmitri Ivanovich Mendeleev	1834–1907	- Used atomic mass to arrange elements - Like Newlands, arranged the elements in periods of 7 - Arranged related elements in columns like Dobereiner's triads - Left gaps for undiscovered elements (e.g. Ge)	- Nationality: Russian - Refused to accept that electrons and radiation and radioactivity existed - Has a radioactive element named after him—mendelevium!
William Ramsay	1852–1916	- Discovered the noble, or inert, gases - Set them as a separate group	- Nationality: Scottish - Received the Nobel Prize in Chemistry
Henry Moseley	1887–1915	- Used atomic number, not mass, to arrange elements - Produced the periodic table still in use today	- Nationality: English - Was shot through the head and killed in World War I at Gallipoli

harmonise (v) to be similar in some way

octave (n) a group of eight notes in music

Use the information in the table and information from Unit 4.2 in your student book to identify:

(a) the scientist who was Scottish _____

(b) the scientist who finished his table before Mendeleev did but received less recognition _____

(c) the scientist who thought that the elements harmonised in octaves like music _____

(d) the scientist who received a Nobel Prize _____

- (e) the scientist who wrote water as H^2O _____
- (f) the scientist who left gaps in his table _____
- (g) the scientist who developed triads _____
- (h) the scientist who thought that light was an element _____
- (i) the scientist who is considered to be the 'father' of the modern periodic table

- (j) the scientist who developed gas laws _____
- (k) the scientist who arranged elements according to their atomic number and not their mass _____
- (l) the scientist who was shot in the head _____
- (m) the scientist who had his head chopped off _____
- (n) the scientist who was Russian _____
- (o) the scientist who revived the idea that atoms existed _____
- (p) the scientist who had 13 000 beakers in his laboratory _____
- (q) the scientist who was put in charge of his school when he was 12

- (r) the scientist who grouped chemicals as organic or inorganic _____
- (s) the scientist who gave the elements the symbols we use today _____
- (t) the scientist who has a radioactive element named after him, even though he didn't believe in radioactivity _____
- (u) the scientist who proved that iron got heavier when it rusted _____
- (v) the scientist who had a statue that should have had its head chopped off because it was the wrong one! _____

TRENDS OF THE PERIODIC TABLE WEB SEARCH

Purpose To investigate some of the characteristics of the periodic table

Use a general search engine such as Dogpile or Metacrawler to find suitable websites that might provide information to answer the questions below. Some useful websites have been listed.

SEARCH: Metacrawler (www.metacrawler.com)

Phrases: (trends of the periodic table), (interactive periodic table)

<http://www.lynchburg.net/hhs/chemistry/trends/>

Click on Cool Periodic Table link.

Quiz on the Table

<http://www.chemcool.com/regents/periodictable/periodictable.htm>

Interactive Table

<http://www.chemicalelements.com/>

<http://www.chemcool.com/>

<http://www.webelements.com/>

<http://www.webelements.com/webelements/scholar/>

<http://www.schoolscience.co.uk/periodictable.html>

Atomic Spectrum

<http://www.colorado.edu/physics/2000/applets/a2.html>

Questions

1. Write down *3 elements (symbols will do)* that belong to the following from the periodic table.
 - (a) Transition metals
 - (b) Alkali metals
 - (c) Non-metals
 - (d) Noble or inert gases
 - (e) Metalloids or semi-metals
 - (f) Halogens
 - (g) Metals
 - (h) Alkali earths
 - (i) Rare earths

2. Define the following as found in the periodic table.
 - (a) Group
 - (b) Period

3. Why are elements placed into the same group in the periodic table?

4. The table below shows the third period elements of the periodic table. Fill in the data required.

	Na	Mg	Al	Si	P	S	Cl	Ar
State of the element at 20 °C.								
Melting point								
Atomic radius								
Electrical conductivity								
Reactivity								

Describe the trends you see.

- (a) State at 20 °C.

- (b) Melting point.

- (c) Atomic radius

- (d) Conductivity

- (e) Reactivity

5. The table below shows Group II elements of the periodic table - *they are metals*. Fill in the data required.

	Be	Mg	Ca	Sr	Ba	Ra
State of the element at 20 °C.						
Melting point						
Atomic radius						
Electrical conductivity						
Reactivity						

Describe the trends you see.

- (a) State at 20 °C.
- (b) Melting point.
- (c) Atomic radius
- (d) Conductivity
- (e) Reactivity

6. The table below shows Group VII elements of the periodic table - *they are non-metals*. Fill in the data required.

	F	Cl	Br	I	At
State of the element at 20 °C.					
Melting point					
Atomic radius					
Electrical conductivity					
Reactivity					

Describe the trends you see.

(a) State at 20 °C.

(b) Melting point.

(c) Atomic radius

(d) Conductivity

(e) Reactivity

Periodic Table

The Periodic Table arranges the elements according to their chemical properties.

Metals are found to the left hand side of the zig-zag on the table. **Non-metals** are found to the right.

Elements on either side of the zig-zag are called **semi-metals** or **metalloids**, showing characteristics of both metals and non-metals.

Hydrogen is a non-metal that behaves like a Group I element sometimes and a Group VII at other times. It is sometimes shown by itself in the table.

All metals with the same chemical properties are arranged in **groups** (vertical columns). They are numbered from I - VIII across the table.

The horizontal rows are called **periods**.

I		II																III	IV	V	VI	VII	VIII	2					
1 H 1.008																													He 4.003
3 Li 6.941	4 Be 9.012															5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18								
11 Na 22.99	12 Mg 24.30	Transition metals														13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95								
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80												
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3												
55 Cs 132.9	56 Ba 137.3	57 *La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)												
87 Fr (223)	88 Ra (226)	89 §Ac (227)	104 Unq (261)	105 Unp (262)	106 Unh (263)	107 Uns (262)	108 Uno (265)	109 Une (266)	110	111																			
		* Lanthanide series		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.2	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0												
		§ Actinide series		90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)												

6

C

12.01

← Atomic number

← Symbol

← Relative atomic mass (atomic weight)

Trends Across the Table

Consider the row 3 elements.

- Go from metals → semi-metals → non-metals.
- Reactivity (ability to react) decreases.

- Conductivity decreases.

i.e. Metals conduct electricity well - non-metals are insulators.

- Melting and boiling points increase from Group I - IV, then decrease from Group V - VIII.

The stronger the bonding between particles, the higher the melting and boiling point.

e.g. H_2O melts at $0\text{ }^\circ\text{C}$, NaCl melts at $801\text{ }^\circ\text{C}$. NaCl has the stronger force between its particles.

Trends Down a Group

- Reactivity increases down a metal group (I-III) - decreases down a non-metal group (IV-VIII).

e.g. Fr (francium) is the most reactive metal (bottom LHS of the table).
F is the most reactive non-metal (top RHS).

- Melting and boiling point decreases down metallic groups (I-III) - increases down non-metal groups.

Special Names of Elements

Alkali Metals

- Group I
- React with H_2O to form alkaline or basic solutions.

Alkaline Earth Metals

- Group II.

Halogens

- Group VII.

Inert (Noble) Gases

- Group VIII.
- Unreactive.
- Exist as single atoms.

Transition Metals

- Found in the middle of the table.
- Elements near each other have similar properties.
- Tend to form coloured compounds with non-metals.

CHEMICAL FAMILIES

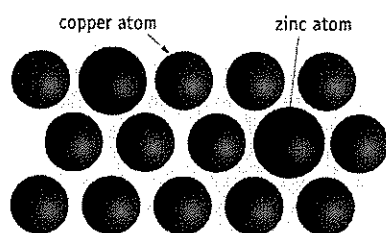
METALS

Properties

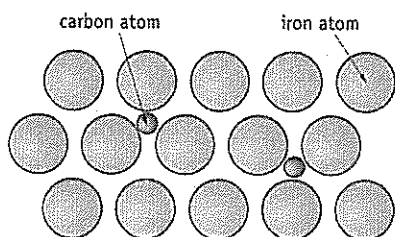
- Good conductors of heat and electricity.
- *Malleable* - can be hammered into sheets.
- *Ductile* - can be drawn/pulled into wires.
- *Metallic lustre* - shiny when polished.

Alloys

- Mixture of two or more metals.
e.g. Silver coins are 75% Cu and 25% Ni.
- Can have atoms of the same size.



- Can have atoms of different sizes.



Reactive Metals

- Group I metals are very reactive (have 1 electron in the outer shell).
- Called *alkali metals*.
e.g. Na in water can explode.
- Group II metals are not as reactive (have 2 electrons in the outer shell).

Transition Metals

- In the middle of the periodic table.
- Hard with high melting points.
- Used to form many alloys.

- Form coloured compounds with non-metals.

e.g. CuSO_4 is blue.

- Cu^{2+} ions are blue.
 Fe^{2+} ions are green.

NON-METALS

Properties

- Found on right hand side of the periodic table.
- Most are gases at room temperature.

Group VII

- Called the *halogens* - very reactive as they have room for one extra electron in their outer shell.
- Form *salts* when combined with metals.

Group VIII

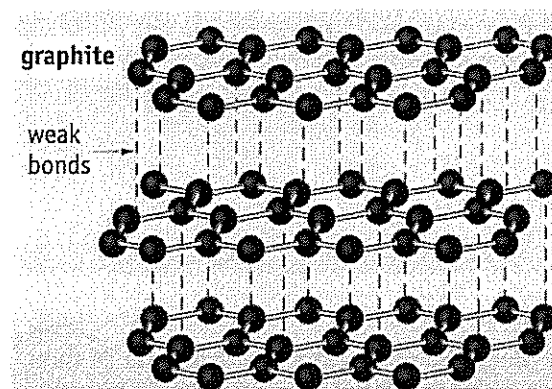
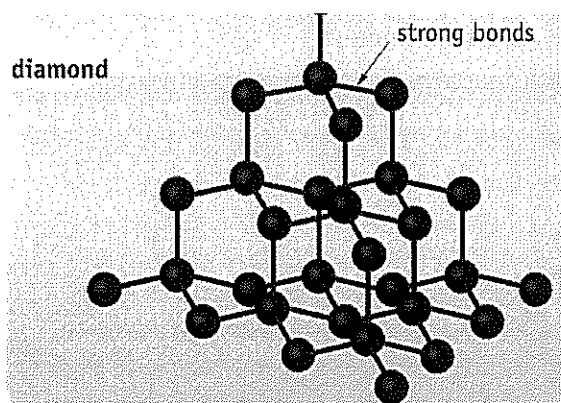
- Are *monatomic* - have one atom only.
- The other non-metal gases are *diatomic* - two atoms together.

e.g. N_2 , O_2 , Cl_2 , F_2 .

Allotropes

- Different forms of the same element.

e.g. Carbon has diamond (hard) and graphite (soft).



Science inquiry

 Visual/Spatial
  Logical/Mathematical

This table shows the melting points, boiling points and atomic radii of the first 20 elements of the periodic table.

radii (n) plural of radius

Element	Atomic number	Melting point (°C)	Boiling point (°C)	Atomic radius (Å)
Hydrogen, H	1	-259	-253	2.1
Helium, He	2	-272	-269	1.9
Lithium, Li	3	181	1347	1.6
Beryllium, Be	4	1278	2970	1.1
Boron, B	5	2300	2550	1.0
Carbon, C	6	3500	4827	0.9
Nitrogen, N	7	-210	-196	0.9
Oxygen, O	8	-218	-183	0.7
Fluorine, F	9	-220	-188	0.6
Neon, Ne	10	-249	-246	0.5
Sodium, Na	11	98	883	1.9
Magnesium, Mg	12	650	1107	1.6
Aluminium, Al	13	660	2467	1.4
Silicon, Si	14	1410	2355	1.3
Phosphorus, P	15	44	280	1.3
Sulfur, S	16	113	445	1.3
Chlorine, Cl	17	-101	-35	1.0
Argon, Ar	18	-189	-186	0.9
Potassium, K	19	64	774	2.4
Calcium, Ca	20	839	1487	2.0

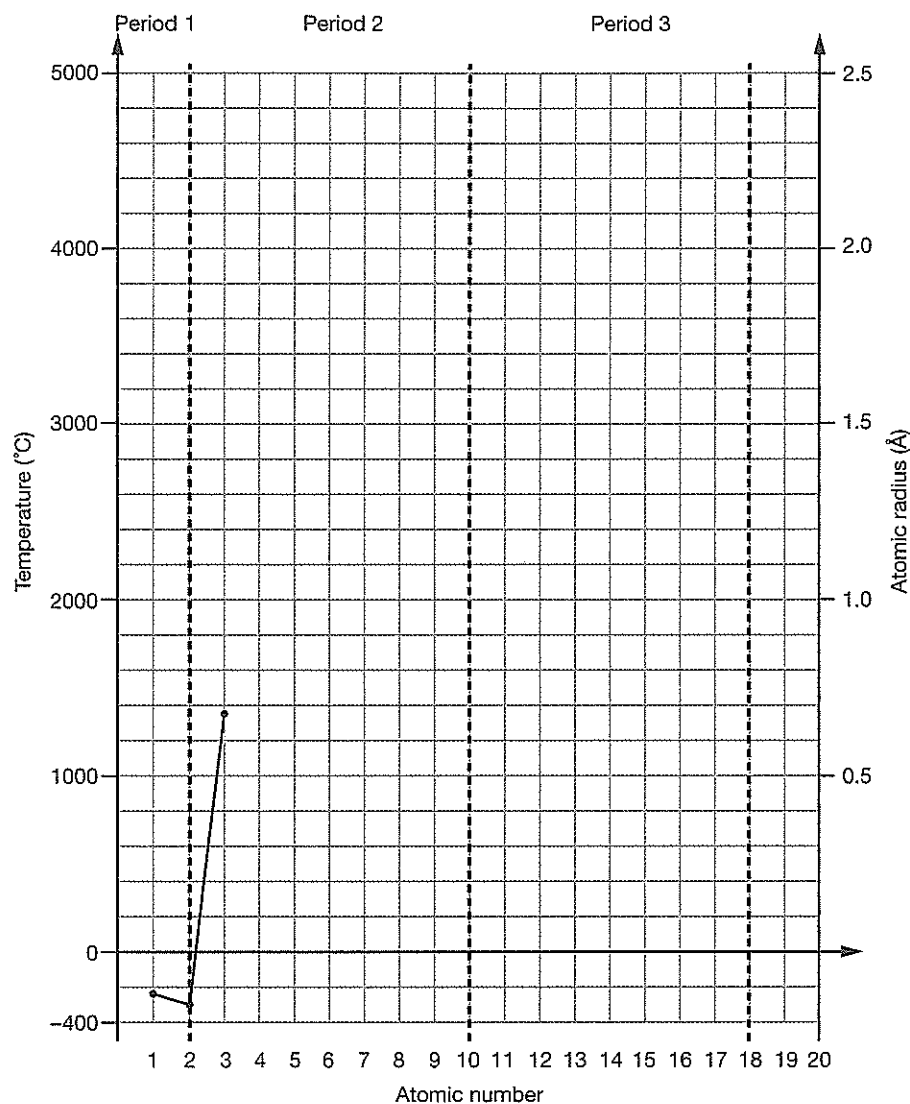
- 1 Use the grid on the next page to **construct** a line graph showing how boiling point varies with atomic number. Use the vertical axis on the left. The graph has been started for you.
- 2 On the same grid, **construct** another line graph showing how melting point varies with atomic number. Use a different colour.
- 3 **Construct** another line graph showing how the atomic radius varies with atomic number. Use the vertical axis on the right of the grid.
- 4 **Compare** the shape of the melting and boiling point graphs.

vertical (adj) in an up and down direction

- 5 **Describe** the overall trend in the melting point, boiling point and atomic radius as you move across a period of the periodic table.

trend (n) direction; pattern

As you move across the period of the periodic table, the melting point _____, the boiling point _____ and the atomic radius _____.



- 6 Describe what happens to the trend in the melting point, boiling point and atomic radius when you move from one period to another.

- 7 Use your graph to **determine** what happens to the size of an atom as you move down a group of the periodic table.

As you move down a group, the atoms _____

- 8 Use the shell structure to **explain** your observations from question 7.

HINT

Look back at worksheet 4.1 to see the shell structure.

Science understanding

Visual/Spatial

1

H

hydrogen

3

Li

lithium

11

Na

sodium

4

Be

beryllium

12

Mg

magnesium

19

K

potassium

20

Ca

calcium

21

Sc

scandium

22

Ti

titanium

23

V

vanadium

24

Cr

chromium

25

Mn

manganese

26

Fe

iron

27

Co

cobalt

28

Ni

nickel

29

Cu

copper

30

Zn

zinc

31

Ga

gallium

32

Ge

germanium

33

As

arsenic

34

Se

selenium

35

Br

bromine

36

Kr

krypton

37

Rb

rubidium

38

Sr

strontium

39

Y

yttrium

40

Zr

zirconium

41

Nb

niobium

42

Mo

molybdenum

43

Tc

technetium

44

Ru

ruthenium

45

Rh

rhodium

46

Pd

palladium

47

Ag

silver

48

Cd

cadmium

49

In

indium

50

Sn

tin

51

Sb

antimony

52

Te

tellurium

53

I

iodine

54

Xe

xenon

55

Cs

caesium

56

Ba

barium

57-71

lanthanides

72

Hf

hafnium

73

Ta

tantalum

74

W

tungsten

75

Re

rhenium

76

Os

osmium

77

Ir

iridium

78

Pt

platinum

79

Au

gold

80

Hg

mercury

81

Tl

thallium

82

Pb

lead

83

Bi

bismuth

84

Po

polonium

85

At

astatine

86

Rn

radon

87

Fr

francium

88

Ra

radium

89-103

actinides

104

Rf

rutherfordium

105

Db

dubnium

106

Sg

seaborgium

107

Bh

bohrium

108

Hs

hassium

109

Mt

meitnerium

110

Ds

darmstadtium

111

Rg

roentgenium

112

Cn

copernicium

113

Uut

ununtrium

114

Uuq

ununquadium

115

Uup

ununpentium

116

Uuh

ununhexium

117

Uus

ununseptium

118

Uuo

ununoctium

13

Al

aluminium

5

B

boron

6

C

carbon

7

N

nitrogen

8

O

oxygen

9

F

fluorine

10

Ne

neon

13

Al

aluminium

14

Si

silicon

15

P

phosphorus

16

S

sulfur

17

Cl

chlorine

18

Ar

argon

1

H

hydrogen

3

Li

lithium

11

Na

sodium

4

Be

beryllium

12

Mg

magnesium

19

K

potassium

37

Rb

rubidium

55

Cs

caesium

20

Ca

calcium

38

Sr

strontium

56

Ba

barium

21

Sc

scandium

39

Y

yttrium

57-71

lanthanides

22

Ti

titanium

40

Zr

zirconium

72

Hf

hafnium

23

V

vanadium

41

Nb

niobium

73

Ta

tantalum

24

Cr

chromium

42

Mo

molybdenum

74

W

tungsten

25

Mn

manganese

43

Tc

technetium

75

Re

rhenium

26

Fe

iron

44

Ru

ruthenium

76

Os

osmium

27

Co

cobalt

45

Rh

rhodium

77

Ir

iridium

28

Ni

nickel

46

Pd

palladium

78

Pt

platinum

29

Cu

copper

47

Ag

silver

79

Au

gold

30

Zn

zinc

48

Cd

cadmium

80

Hg

mercury

31

Ga

gallium

49

In

indium

81

Tl

thallium

32

Ge

germanium

50

Sn

tin

82

Pb

lead

33

As

arsenic

51

Sb

antimony

83

Bi

bismuth

34

Se

selenium

52

Te

tellurium

84

Po

polonium

35

Br

bromine

53

I

iodine

85

At

astatine

36

Kr

krypton

54

Xe

xenon

86

Rn

radon

87

Fr

francium

104

Rf

rutherfordium

106

Sg

seaborgium

108

Hs

hassium

110

Ds

darmstadtium

112

Cn

copernicium

114

Uuq

ununquadium

116

Uuh

ununhexium

118

Uuo

ununoctium

88

Ra

radium

105

Db

dubnium

107

Bh

bohrium

109

Mt

meitnerium

111

Rg

roentgenium

113

Uut

ununtrium

115

Uup

ununpentium

117

Uus

ununseptium

89-103

actinides

57

La

lanthanum

72

Hf

hafnium

58

Ce

cerium

73

Ta

tantalum

59

Pr

praseodymium

74

W

tungsten

60

Nd

neodymium

75

Re

rhenium

61

Pm

promethium

76

Os

osmium

62

Sm

samarium

77

Ir

iridium

63

Eu

europium

78

Pt

platinum

64

Gd

gadolinium

79

Au

gold

65

Tb

terbium

80

Hg

mercury

66

Dy

dysprosium

81

Tl

thallium

67

Ho

holmium

82

Pb

lead

68

Er

erbium

83

Bi

bismuth

69

Tm

thulium

84

Po

polonium

70

Yb

ytterbium

85

At

astatine

71

Lu

lutetium

86

Rn

radon

89

Ac

actinium

104

Rf

rutherfordium

106

Sg

seaborgium

108

Hs

hassium

110

Ds

darmstadtium

112

Cn

copernicium

114

Uuq

ununquadium

116

Uuh

ununhexium

118

Uuo

ununoctium

90

Th

thorium

105

Db

dubnium

107

Bh

bohrium

109

Mt

meitnerium

111

Rg

roentgenium

113

Uut

ununtrium

115

Uup

ununpentium

117

Uus

ununseptium

91

Pa

protactinium

106

Sg

seaborgium

108

Hs

hassium

110

Ds

darmstadtium

112

Cn

copernicium

114

Uuq

ununquadium

116

Uuh

ununhexium

118

Uuo

ununoctium

92

U

uranium

107

Bh

bohrium

109

Mt

meitnerium

111

Rg

roentgenium

113

Uut

ununtrium

115

Uup

ununpentium

117

Uus

ununseptium

93

Np

neptunium

108

Hs

hassium

110

Ds

darmstadtium

112

Cn

copernicium

114

Uuq

ununquadium

116

Uuh

ununhexium

118

Uuo

ununoctium

94

Pu

plutonium

109

Mt

meitnerium

111

Rg

roentgenium

113

Uut

ununtrium

115

Uup

ununpentium

117

Uus

ununseptium

95

Am

americium

110

Ds

darmstadtium

112

Cn

copernicium

114

Uuq

ununquadium

116

Uuh

ununhexium

118

Uuo

ununoctium

96

Cm

curium

111

Rg

roentgenium

113

Uut

ununtrium

115

Uup

ununpentium

117

Uus

ununseptium

97

Bk

berkelium

112

Cn

copernicium

114

Uuq

ununquadium

116

Uuh

ununhexium

118

Uuo

ununoctium

98

Cf

californium

113

Uut

ununtrium

115

Uup

ununpentium

117

Uus

ununseptium

99

Es

einsteinium

114

Uuq

ununquadium

116

Uuh

ununhexium

118

Uuo

ununoctium

100

Fm

fermium

115

Uup

ununpentium

117

Uus

ununseptium

101

Md

mendelevium

116

Uuh

ununhexium

102

No

nobelium

103

Lr

lawrencium

1

H

hydrogen

3

Li

lithium

11

Na

sodium

4

Be

beryllium

12

Mg

magnesium

19

K

potassium

37

Rb

rubidium

55

Cs

caesium

20

Ca

calcium

38

Sr

strontium

56

Ba

barium

21

Sc

scandium

39

Y

yttrium

57-71

lanthanides

22

Ti

titanium

40

Zr

zirconium

72

Hf

hafnium

23

V

vanadium

41

Nb

niobium

73

Ta

tantalum

24

Cr

chromium

42

Mo

molybdenum

74

W

tungsten

25

Mn

manganese

43

Tc

technetium

75

Re

rhenium

26

Fe

iron

44

Ru

ruthenium

76

Os

osmium

27

Co

cobalt

45

Rh

rhodium

77

Ir

iridium

28

Ni

nickel

46

Pd

palladium

78

Pt

platinum

29

Cu

copper

47

Ag

silver

79

Au

gold

30

Zn

zinc

48

Cd

cadmium

80

Hg

mercury

31

Ga

gallium

49

In

indium

81

Tl

thallium

32

Ge

germanium

50

Sn

tin

82

Pb

lead

33

As

arsenic

51

Sb

antimony

83

Bi

bismuth

34

Se

selenium

52

Te

tellurium

84

Po

polonium

35

Br

bromine

53

I

iodine

85

At

astatine

36

Kr

krypton

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Ce

cerium

73

Ta

tantalum

59

Pr

praseodymium

74

W

tungsten

60

Nd

neodymium

75

Re

rhenium

61

Pm

promethium

76

Os

osmium

62

Sm

samarium

77

Ir

iridium

63

Eu

europium

78

Pt

platinum

64

Gd

gadolinium

79

Au

gold

65

Tb

terbium

80

Hg

mercury

66

Dy

dysprosium

81

Tl

thallium

67

Ho

holmium

82

Pb

lead

68

Er

erbium

83

Bi

bismuth

69

Tm

thulium

1 Use different colours or shading to **demonstrate** where the following elements are on the simple version of the periodic table shown above. Note that some of your coloured or shaded areas might overlap.

- noble or inert gases
- alkali metals
- actinides
- transition elements
- halogens
- alkaline earths
- lanthanides

2 Circle an element symbol to **identify**:

- the only non-metal that exists as a liquid at normal room temperature
- the element found in all living (and dead) things, being the basis for most of their molecules
- the lightest of all elements
- the heaviest of all elements.

overlap (v) for one thing to partly cover another